A Low-Rank Perspective on Similarity Matrix Completion

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***Abstract*—In real-world information retrieval scenarios, addressing data incompleteness is crucial for providing accurate similarity scores and ensuring reliable results for downstream tasks. Previous Similarity Matrix Completion (SMC) methods aim to estimate a similarity matrix that exhibits positive semi-definiteness (PSD) from an inaccurate one. However, these methods are inadequate in cases where the initial similarity matrix S^0 exhibits PSD. In this paper, we propose a novel SMC framework that simultaneously explores the symmetric, positive semi-definiteness (PSD), and low-rank properties to provide an accurate and efficient solution for similarity search tasks. Specifically, we exploit the low-rank property and introduce an efficient specialized Cholesky factorization (CF) technique into the conventional SMC framework, which is implemented via a regularizer. It improves computational efficiency by learning a smaller factorized matrix instead of the entire similarity matrix. Moreover, to enhance the optimality guarantees of SMC, we introduce a novel lower-rank matrix property and design two corresponding regularizers. Building upon these meticulous designs, our novel SMC framework, for the first time, ensures both efficiency and accuracy with theoretical guarantees. Consequently, we propose two novel algorithms, i.e., SMCFN/SMCRN, to implement the SMC framework. Theoretical analysis verifies the effectiveness and fast convergence speed of SMCFN/SMCRN. Extensive experiments on five real-world datasets validate our theoretical analysis: SMCFN/SMCRN achieves up to 42% lower RMSE and 15% higher Recall than state-of-the-art baselines, while being the most efficient among all baseline methods.**

***Index Terms*—Data Incompleteness, similarity matrix completion.**

I. Introduction

IN REAL-WORLD information retrieval scenarios, data incompleteness is ubiquitous due to partial observations. This poses a prevalent challenge to calculating accurate similarity scores and adversely degrades the performance of downstream retrieval tasks. For example, images might be incompletely observed due to occlusion or corruption [1], thus resulting in inaccurate image retrieval results. Similarly, documents may be partially corrupted due to scanning errors or file corruption [2], thus resulting in inaccurate text retrieval results.

Given a data matrix $X \in \mathbb{R}^{d\times n}$, denoting n data samples in d-dimensional space, the correlated similarity matrix $S^* \in \mathbb{R}^{n\times n}$ denotes the pairwise similarity scores of n data samples. Specifically, each entry S^*_{ij} is calculated by a specific similarity metric, with the symmetric similarity metric [3] being widely adopted in practical applications [4], [5], [6]. This ensures that S^* naturally exhibits symmetric (i.e., $S = S^\top$) and positive semi-definiteness (PSD) properties (i.e., $S \succeq 0$) [7], [8], [9], where all the eigenvalues are non-negative [9], [10].

However, due to data incompleteness, where the unobserved entries in the data matrix X^0 are missing at random [11], [12], [13], the initial similarity matrix S^0 derived from a partially observed data matrix X^0 may not accurately represent the true pairwise similarity scores. Specifically, S^0 is computed based on the available entries in X^0, using the predefined similarity metric that operates on these incomplete observations. This process may introduce inaccuracies since the similarity are calculated with the partially observed information, which potentially leads to an inaccurate S^0 that does not sufficiently approximate the true similarity scores, thereby degrading the performance of the downstream tasks relying on S^0.

To address this issue and improve the efficacy of downstream retrieval tasks, similarity matrix completion (SMC) [14], [15], [16], [17] arises as a straightforward approach, which aims to develop reliable, efficient, and scalable algorithms to identify a solution $\hat{S}$ that closely approximates the ground truth S^* while satisfying other prior constraints.

Previous SMC methods typically leverage the PSD property to ensure that the estimated similarity matrix effectively calibrates pairwise similarities [7], [8], [9]. This method is crucial for maintaining a reliable similarity matrix, which forms the basis for various downstream retrieval tasks. To enforce the PSD constraint, these methods employ singular value decomposition (SVD) operation [18], [19] to extract the underlying non-negative eigenvalues as guidance for the SMC process. However, performing a full SVD on a similarity matrix $S \in \mathbb{R}^{n\times n}$ scales with the computation cost $O(n^3)$ and storage cost $O(n^2)$ [18], [19], [20], thereby limiting the efficiency of conventional SMC methods in practical large-scale applications.

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Moreover, in cases where the initial similarity matrix S^0 exhibits PSD, the optimal solution of SMC is exactly S^0, which achieves a minimum objective value of 0. This implies that conventional SMC algorithms cannot provide any enhanced solution over the initial similarity matrix S^0 when it already satisfies the PSD constraint. Thus, alternative constraints or strategies are necessitated, such as incorporating low-rank properties or advanced regularization techniques, to enhance the accuracy and reliability of SMC.

A. Novel Similarity Matrix Completion Framework

In response, we investigate the low-rank property of the similarity matrix, arising from high correlations among rows or columns, as noted in [21]. Thus, we formulate a novel SMC framework that integrates low-rank and PSD properties, which is expressed as follows:

$$\min_{S} \|S - S^0\|_F^2, \text{ s.t. } \mathrm{S} \succeq 0, \text{ rank(S)} < \mathrm{r}. \tag{1}$$

Here, similarity matrix $S \in \mathbb{R}^{n\times n}$ represents the pairwise similarities, where n denotes the number of data samples, each element S_{ij} denotes the similarity score of pairwise data samples, and $\|.\|_F^2$ denotes the Frobenius norm. $\text{rank}(S) < r$ denotes the low-rank property of the similarity matrix S, and $S \succeq 0$ denotes a real symmetric similarity matrix S that exhibits the PSD property, which always holds when any symmetric similarity metric is adopted. By adopting the newly obtained low-rank property, the novel SMC framework can avoid the special case where the initial similarity matrix S^0 is the optimal solution.

To enforce the low-rank constraint, the widely used method is to replace the low-rank constraint by minimizing the nuclear norm $\|S\|_*$ as the surrogate [22]. Specifically, calculating the nuclear norm involves the summation of the eigenvalues of the similarity matrix S, which is achieved by SVD operation [18], [19], [20]. However, the computational and storage costs associated with performing SVD constitute the primary barriers to applying it to large-scale problems.

B. Efficient Matrix Factorization Technique for PSD Matrix

As a technique investigated in recent decades, matrix factorization (MF) [23], [24] is utilized to improve computational efficiency on a large scale [25], [26], [27], [28]. Specifically, MF learns two factorized matrices with a rank smaller than that of the entire large matrix [29], [30]. For instance, applying the MF on similarity matrix $S \in \mathbb{R}^{n\times n}$ involves decomposing it into a smaller factorized matrix $V \in \mathbb{R}^{n\times r}$ $(r < n)$, which results in affordable computational cost $O(n^2 r)$ and a low storage cost $O(nr)$. Furthermore, it enables scalability to problems of large sizes when learning-based optimization methods are adopted [31].

Despite extensive adoption and impressive performance of MF techniques in practical applications [32], [33], [34], their nonconvex structure [23], [24] poses a significant challenge. Specifically, the foundational understanding of generic nonconvex optimization remains underdeveloped, which often results in convergence to a local minimum rather than to the desired global solution, thus compromising optimality. Theoretical foundations for problems involving matrix factorization (MF) techniques have been insufficient until very recently, given that these problems are inherently nonconvex. This deficiency has been a significant challenge, especially in ensuring that optimization methods consistently achieve global optima in real-world scenarios. To address this challenge, recent studies [35], [36], [37] present theoretical optimality guarantees by incorporating a regularizer with the MF technique in low-rank matrix completion (MC) problems. Inspired by these findings, we investigate whether the global optimality guarantee extends to the SMC framework: *Can the global optimality guarantee be extended to the SMC framework?*

C. Contributions

To address the above problems, this paper aims to provide a thorough overview of SMC methods from the low-rank perspective, targeting the broader foundational machine learning, information retrieval, and optimization communities. Our contributions are summarized in three points:

- *Novel SMC Framework Achieves Efficient Solution:* We introduce a novel SMC framework that leverages the *symmetric*, *positive semi-definite (PSD)*, and *low-rank* properties of the similarity matrix. On the one hand, we exploit the symmetric and PSD properties and consequently introduce a tailored Cholesky factorization (CF) [38] technique into the SMC framework to improve computational efficiency. On the other hand, we introduce the *low-rank* property and enforce the accuracy via the newly proposed lower-rank matrix property. Therefore, our SMC framework learns only the smaller factorized matrix rather than the entire similarity matrix, significantly accelerating the SMC process with an effective solution.
- *Enhancing the Optimality Guarantee:* Moreover, we propose a novel lower-rank matrix property to further provide an optimality guarantee of the SMC process. Specifically, it ensures that the factorized matrix learned by Cholesky Factorization (CF) can accurately reconstruct the similarity matrix while strictly preserving the low-rank property. To this end, we design two regularizers that can be seamlessly incorporated into the SMC framework. One regularizer minimizes the rank of the factorized matrix by employing a tight upper-bound Frobenius norm of the low-rank factorized matrix, serving as a surrogate for minimizing the rank of the similarity matrix, avoiding the computationally intensive rank minimization function. The other regularizer further ensures the optimality guarantee by imposing a stricter constraint on the Frobenius norm of each row vector in the factorized matrix. Accordingly, we propose two general and efficient algorithms, SMCFN and SMCRN, to implement the proposed SMC framework.
- *Superior Performance:* We conduct extensive experiments on various real-world datasets. Comparative analysis consistently shows that our SMCFN/SMCRN algorithm outperforms the comparison methods on similarity matrix

completion tasks. Importantly, our algorithms achieve these enhanced results while maintaining efficiency, highlighting the practical viability for real-world applications.

D. Comparison of Existing Methods

Existing methods for SMC and related problems present the following limitations:

- *PSD-based SMC methods:* [14], [15], [39]: These methods enforce PSD property via SVD with $O(n^3)$ complexity while disregarding the low-rank property.
- *Low-rank MC methods:* [30], [40], [41], [42]: These methods assume entries are missing at random (MAR), which does not hold for SMC (see Section II-C). Moreover, they do not guarantee PSD.

We propose SMCFN/SMCRN, which is the first SMC framework that *jointly* enforces PSD and low-rank properties, thus achieving a provable convergence rate of $O(1/T)$ and global optimality guarantees.

E. Notation

Lowercase letters represent vectors, and uppercase letters represent matrices. $(.)^\top$ denotes the transpose operation, and I denotes the identity matrix. A square matrix $X \in \mathbb{R}^{n\times n}$, $\mathrm{tr}(X)$ is its trace, $\|X\|_F = \sqrt{\mathrm{tr}(X^\top X)}$ is its Frobenius norm. $\|X\|_\infty = \max_{1\le i\le n}\sum_{j=1}^{n}|X_{ij}|$ is its infinity norm. The singular value decomposition (SVD) [19] on a symmetric positive semi-definite matrix X is $V\Sigma V^\top$, where $V \in \mathbb{R}^{n\times n}$, $\Sigma = diag(\sigma(X)) = [\sigma_i(X)]$ with $\sigma_i(X)$ being the ith singular value of X and w.l.o.g., $\sigma_1(X) \ge \sigma_2(X) \ge \ldots \ge \sigma_n(X) \ge 0$.

II. Background

We first introduce the calculation of the similarity matrix with data incompleteness. Subsequently, we discuss the related studies for addressing the data-incompleteness problem. Next, we introduce the definition of the similarity matrix completion (SMC) problem and the prior methods, along with the necessary prerequisites. Ultimately, we discuss the SMC setting and explain the differences with respect to other related areas.

A. Similarity Matrix Calculation

Consider a search dataset of items from which we aim to retrieve relevant entries, denoted as $P = \{p_1,\ldots p_n\} \in \mathbb{R}^{d\times n_p}$, where n_p denotes the number of data samples in d-dimensional space. For example, p_i denotes a d-dimensional data vector of the ith item, such as an image, text, or video. Given a query vector $q \in \mathbb{R}^d$from the query database $Q \in \mathbb{R}^{d\times n_q}$, the retrieval task aims to identify the top-k most similar candidates from the reference database $P \in \mathbb{R}^{d\times n_p}$, based on a similarity metric κ: $\mathbb{R}^d \times \mathbb{R}^d \to \mathbb{R}$. Following the paradigm established in real-world applications [43], [44], [45], we model the pairwise relationships between P and Q via a global similarity matrix $X \in \mathbb{R}^{n\times n}$ (where $n = n_p + n_q$), which is constructed from the union of these two vector databases [46]. We adopt the Cosine similarity [47] as the similarity metric κ to measure the similarities between pairwise data vectors. Ideally, we aim to receive k retrieval items $\mathcal{R}(q) = \{p_1,\ldots,p_k\} \in P$ such that $\kappa(o,p_i) \ge \kappa(o,p_j)$ for all $p \in P$ and $i \in [k]$, where $i \ne j$.

Given the initial data matrix X^0, the initial similarity matrix S^0 can be calculated by cosine similarity [47], formulated as:

$$S^0_{ij} = \frac{\mathcal{P}_\Omega(X^0_i)^\top \mathcal{P}_\Omega(X^0_j)}{\|\mathcal{P}_\Omega(X^0_i)^\top\|\|\mathcal{P}_\Omega(X^0_j)\|}, \tag{2}$$

where $\Omega = \{(i,j)|X_{i,j} \text{ is given}\}$ is the set of indices of the positions of the observed entries in the initial data matrix X^0. $\|.\|_F$ is the Frobenius norm of a matrix, $\mathcal{P}_\Omega$ is the orthogonal sampling operator, which is expressed as:

$$[\mathcal{P}_\Omega(X^0)]_{ij} = \begin{cases} X^0_{ij} & \text{if } \Omega_{ij} = 1,\\ 0 & \text{otherwise.}\end{cases}$$

Here, $\Omega_{ij} = 1$ if X^0_{ij} is observed, and $\Omega_{ij} = 0$ otherwise. When the data matrix X is fully observed, the similarity matrix S^* measures the pairwise similarities of data samples in data matrix X^*, which is exactly the ground truth.

B. Related Work of Handling Data Incompleteness

Data incompleteness poses a significant challenge in machine learning applications and typically results in an inaccurate similarity matrix S^0 when calculated from an incomplete data matrix X^0 using (2). Various methods address this issue by directly completing the matrix X^0 to obtain a complete matrix $\hat{X}$, which in turn leads to a more accurate and reliable similarity matrix $\hat{S}$, thereby improving the accuracy of the initial similarity matrix S^0.

1) Missing Data Imputation: Given an incomplete data matrix $X^0 \in \mathbb{R}^{d\times n}$, where each column X_i denotes a data sample with incomplete observations. The straightforward approach is to directly substitute the missing values with alternative values in the incomplete raw data sample X_i. For example, single-value imputation techniques, i.e., mean value imputation [48] or zero imputation [49] methods are easy to implement but may not fully account for uncertainties. Moreover, model-based methods such as Bayesian methods [50], linear regression [51], and maximum likelihood [52] provide a statistical approach by modeling the relationship between the missing values and observed values. However, the missing data substitution methods may change the data variability or incorrectly capture relationships between data samples [53].

2) Data Matrix Completion: Another approach to handling data incompleteness is data matrix completion (MC) [41], [54], [55], [56], which leverages the specific low-rank property estimate of the incomplete observed data matrix $\hat{X}$, by substituting the missing data in the raw incomplete data matrix X^0 and making $\hat{X}$ approximate the unknown complete matrix X^* [25], [26], [27], [28]. The objective function is formulated as $\min_X \|X - X^0\|_F^2$ under a low-rank constraint, ensuring $\mathrm{rank}(X) \le r$ on the estimated matrix $\hat{X}$.

Under reasonable conditions [57], [58], [59], the rank constraint $\mathrm{rank}(\cdot)$ can be effectively managed by solving the surrogate enjoys a strong theoretical guarantee for exact matrix estimation [41], specifically the nuclear norm $\|X\|_*$, rather than directly minimizing rank(X). One category is the fixed-point shrinkage algorithm [60], which addresses the regularized linear least squares problem. The other is iterative SVD [61], [62], which involves selecting singular values of the partially observed matrix in each iteration. However, both methods require singular value decomposition (SVD), which introduces high computational costs that escalate with the size or rank of the matrix.

C. Similarity Matrix Completion (SMC)

SMC problem is defined and formulated in problem (1). Given the initial inaccurate similarity matrix S^0, SMC aims to find the optimal estimated $\hat{S}$ that satisfies the constraints as the solution to approximate the ground truth S^*.

Prior SMC methods [14], [15], [16], [17] primarily focus on the symmetric and PSD properties of the similarity matrix while solely ignoring the low-rank property. These methods typically leverage the PSD property and employ singular value decomposition (SVD) operation [19] to extract underlying information through non-negative eigenvalues as well as ensure the PSD property of the estimated similarity matrix. However, these methods may encounter computational bottlenecks when performing SVD operations on large-scale datasets due to time complexity $O(n^3)$ and storage requirements $O(n^2)$. To mitigate the computational cost, a learning-based method [39] is proposed to perform a one-time SVD operation on a fully observed subset of the similarity matrix to find the eigenvalues as guidance. Despite this approach partially reducing computational load, the SVD operation remains a substantial burden, particularly for extremely large datasets.

Moreover, in certain cases where the initial similarity matrix exhibits PSD, these SMC methods may simply return the initial similarity matrix without providing any improvement. Therefore, our proposed novel low-rank SMC framework arises as an effective and efficient choice for SMC problems. It further incorporates the low-rank property into the SMC framework that handles the large-scale dataset and addresses scenarios where the initial matrix is already PSD.

1) Discussion: A similar yet different method is the MC method that provides a solution to recover the real complete matrix [41], [54], [55], [56]. Mathematically, given an incomplete data matrix X^0 where the unobserved entries are missing at random, the correlated initial inaccurate similarity matrix defined as $S'^0 = \mathcal{P}_{\Omega'}(S^*)$, where

$$[\mathcal{P}_{\Omega'}(S'^0)]_{ij} = \begin{cases} S^*_{ij} & \text{if } \Omega'_{ij} = 1, \\ S'^0_{ij} & \text{otherwise.} \end{cases}$$

Here, $\Omega' \in \{0,1\}^{n\times n}$ indicates the positions of observed entities in the initial similarity matrix S'^0. The MC methods can be expressed as:

$$\min_S \|S - S'^0\|_F^2 \quad \text{s.t. S} \succeq 0, \text{rank(S)} < \text{r}. \tag{3}$$

Problem (3) addresses the estimation of a low-rank and positive semi-definite (PSD) similarity matrix S from partially observed entries S'^0.

TABLE I
RANK(S^*), WHERE $S^* \in \mathbb{R}^{n\times n}$, AND $n = n_p + n_q$. WE RANDOMLY SAMPLE $n_p = 1,000, 2,000, 3,000$ SEARCH CANDIDATES AND $n_q = 200$ QUERY ITEMS

Datasets	rank($S^*_{n=1,200}$)	rank($S^*_{n=2,200}$)	rank($S^*_{n=3,200}$)
ImageNet	978	992	1000
MNIST	484	501	515
PROTEIN	357	357	357
CIFAR-10	501	512	512
GoogleNews	300	300	300

It is unsuitable for the similarity matrix completion (SMC) scenario. Specifically, combined with the similarity matrix calculation in (2), SMC problem (1) can be reformulated to:

$$\min_S \|S - P_\Omega(M)^\top \mathcal{P}_\Omega - (X_i^0)^\top \mathcal{P}_\Omega(X_j^0) P_\Omega(M)\|_F^2$$
$$\text{s.t. S} \succeq 0, \text{ rank(S)} < \text{r}.$$

Where $P_\Omega(M) = \text{diag}(\|P_\Omega(x_1^0)\|, \|P_\Omega(x_2^0)\|, \ldots, \|P_\Omega(x_n^0)\|)$, x_i^0 denotes its i-th column of X^0.

Clearly, while the entries in X^0 are missing at random, the missing schema in S'^0 cannot be ensured to be missing at random. This is the fundamental difference between the Matrix Completion (MC) problem and the Similarity Matrix Completion (SMC) problem.

III. METHODOLOGY

In this section, we first introduce the formulation for our novel low-rank SMC framework. Subsequently, to improve efficiency, we introduce the newly proposed lower-rank matrix property via the MF technique into the SMC framework, enabling learning a small factorized matrix rather than the entire matrix while enforcing effectiveness. Next, we incorporate the lower-rank matrix property into the SMC framework through two regularizers and propose two general and efficient algorithms accordingly.

A. Low Rank SMC Framework

We propose a novel low-rank SMC framework, which not only investigates the symmetric and positive semi-definite (PSD) properties as in previous work but also exploits the low-rank matrix property. As we discussed in Section I, the low-rank property of the similarity matrix arises from the high correlation among its rows/columns [21], which is different from the low-rank property in MC methods that comes from the incomplete observed data samples. As also shown in [28], when data exhibits clustering patterns, the similarity matrix becomes approximately block-diagonal, where each diagonal block corresponds to within-cluster similarities. Each block can be approximated by a low-rank outer product, making the entire matrix low-rank with rank approximately equal to the number of clusters.

Table I verifies that the low-rank property of $S^* \in \mathbb{R}^{n\times n}$ holds for various datasets with different sizes.

Based on this insight, we formulate the novel SMC framework as shown in (1) that simultaneously incorporates symmetric, PSD, and the newly found low-rank properties to find the estimated solution $\hat{S}$ that approximates the ground truth S^*.

1) Similarity Matrix Estimation Guarantee: Theorem 3.1 provides a theoretical guarantee that the optimal estimated similarity matrix $\hat{S}$ found by problem (1) is closer to or at least identical to (in special cases) the fully observed ground-truth similarity matrix S^* than the initial similarity matrix S^0.

Theorem 3.1 (Estimation Guarantee): $||S^* - \hat{S}||_F^2 \leq ||S^* - S^0||_F^2$. The equality holds if and only if $\hat{S} = S^0$.

Proof:

$$\begin{aligned} & ||S^* - \hat{S}||_F^2, \\ \leq\ & ||S^* - \hat{S}||_F^2 - 2\langle S^* - \hat{S}, S^0 - \hat{S}\rangle, \\ \leq\ & ||S^* - \hat{S}||_F^2 + ||S^0 - \hat{S}||_F^2 - 2\langle S^* - \hat{S}, S^0 - \hat{S}\rangle, \\ =\ & ||(S^* - \hat{S}) - (S^0 - \hat{S})||_F^2, \\ =\ & ||S^* - S^0||_F^2. \end{aligned}$$

Note that $\hat{S} = \mathcal{P}_C(S^0)$, where $C \subseteq \mathbb{R}^{n\times n}$ is the nonempty closed convex set of low-rank and PSD algebraic variety, and $\mathcal{P}$ is the Euclidean projection onto C. Then, $\langle S^* - \hat{S}, S^0 - \hat{S}\rangle \leq 0$, by the various characterization of convex projection refer to Theorem 4.3-1 [63]. □

Remark: Theorem 3.1 provides a theoretical guarantee to ensure the estimated similarity matrix $\hat{S}$ always approximates the ground-truth similarity matrix S^*.

B. Efficient Lower Rank Matrix Property Enforce Optimality Guarantee

As discussed in Section I, previous work enforces the PSD property via the SVD operation to ensure the eigenvalues are non-negative, which exhibits $O(n^3)$ computational complexity and $O(n^2)$ storage complexity when applied to a similarity matrix $S \in \mathbb{R}^{n\times n}$. The high complexity imposes a significant computational burden, particularly for large-scale datasets. To improve the computational efficiency, we introduce the matrix factorization (MF) technique [23], [24] into the SMC framework. A key insight is that any PSD matrix $S \succeq 0$ can be expressed as $S = VV^\top$ for some matrix V. This is a generalized form of Cholesky factorization [36], [38]. Unlike the standard Cholesky decomposition that factorizes a *known* PSD matrix into lower triangular factors, we use this factorization to *parameterize* the unknown similarity matrix. This approach offers an advantage: the PSD constraint is automatically satisfied for any V, eliminating the need for expensive eigenvalue projections.

Therefore, problem (1) involving the CF technique can be reformulated as:

$$\min_V \|VV^\top - S^0\|_F^2. \quad \text{s.t.}\ \mathrm{rank(VV^\top) < r,\ V \in \mathbb{R}^{n\times r}}, \tag{4}$$

where the PSD property $S \succeq 0$ is enforced by the CF technique mechanically, shown as $\mathrm{rank}(VV^\top) < r$. However, the problem (4) is non-convex and poses challenges in ensuring the convergence of the optimal and unique solution [23], [24], which means it may easily converge to a local optimum.

To mitigate these issues, we first use a mild assumption for our theoretical analysis:

Assumption 3.2: $||V||_F \leq G_1$, $||V||_\infty \leq G_2$.

Here, G_1 and G_2 are constants. This assumption is based on a previous study [64], which indicates the norm of the matrix is bounded.

The following theorem shows that all local optimums would be global optimums when $rank(V'V'^\top) < r$.

Theorem 3.3 (No spurious local minima): Any stationary point V' with $rank(V'V'^\top) < r$ to problem (4) is also the global optimum.

Proof:

Recall that the general convex Semi-Definite Programming (SDP) [65] problem can be stated as:

$$\begin{array}{ll} \min_S & f(S) \\ \text{s.t.} & \mathrm{tr}\,(A_i S) = b_i, A_i \in \mathbb{S}^n, b_i \in \mathbb{R}, i = 1, \dots, m \\ & S \succeq 0 \end{array} \tag{5}$$

Taking $\forall i, A_i = 0, b_i = 0$, and $f(S) = \|S - S^0\|_F^2$, we have:

$$\min_S \|S - S^0\|_F^2 \quad \text{s.t. } \mathrm{S} \succeq 0, \tag{6}$$

which is exactly problem (1) without the rank constraint. According to Theorem 6 in [65], the stationary point in problem (5) with low-rank constraint ($\mathrm{rank}(VV^\top) < r$) is the optimal solution for problem (5). Therefore, the local optimum solution to problem (4) is also the optimal solution for problem (1). □

Theorem 3.3 asserts that when the rank of the factorized matrix V matches the intrinsic rank r of the original similarity matrix S, any stationary point V' that satisfies $\mathrm{rank}(V'V'^\top) < r$ must be a global optimum.

Next, we prove that the stationary point V' with $\mathrm{rank}(\hat{S}) = \mathrm{rank}(V'V'^\top) < r$, where $\|\hat{S} - S^*\| = \|V'V'^\top - S^*\|$ is bounded under a mild assumption:[1]

Theorem 3.4: [Bounded Guarantee] Let $S^* \in \mathbb{R}^{n\times n}$ be the ground-truth similarity matrix and $p = |\Omega|/n^2$ be the observation ratio. The optimal solution to problem (4), denoted as $\hat{S}$, satisfies the bound with high probability:

$$\|\hat{S} - S^*\| \leq \sigma_1^2 r \left\| O\left(G_2^2 nr \log n + \sqrt{pNrG_2^2 G_3^2 \log n}\right)\right\|^2 + pn^2.$$

1) Smooth Relaxation of Lower-Rank Constraint: To solve the problem (4) smoothly [57], [66], [67], [68], [69], [70], we incorporate the hard low-rank constraint $\mathrm{rank}(VV^\top) < r$ into the objective function as a regularizer $R(V) = \mathrm{rank}(VV^\top)$, weighted by a parameter $\lambda > 0$. Now, the problem (4) is converted to:

$$\min_V \|VV^\top - S^0\|_F^2 + \lambda \mathrm{rank}(VV^\top) \quad \text{s.t.}\ \mathrm{V \in \mathbb{R}^{n\times r}}, \tag{7}$$

where the PSD property is ensured by the low-rank CF technique mechanically, shown as $\mathrm{rank}(VV^\top) < r$, hard low-rank constraint is relaxed as the regularizer $R(VV^\top) < r$, and lower-rank matrix property of $\hat{S}$ is ensured by enforcing $V \in \mathbb{R}^{n\times r}$. In other

[1] All proofs are in Supplemental material

words, problem (7) aims to balance the goodness of estimation via the squared loss (expressed as the first term) and the low-rank property of the matrix (represented as the second term).

However, solving problem (7) is still computationally intractable with the rank function as a regularizer $R(V) = \text{rank}(VV^\top)$ [22], [27], [70], which means it is non-trivial to solve all instances optimally within polynomial time. The nuclear norm [27], [41], [70], denoted as $R(V) = \|VV^\top\|_*$, can be utilized as a surrogate [41] to obtain a computationally feasible solution, formulated as:

$$\min_V \|VV^\top - S^0\|_F^2 + \|VV^\top\|_* \quad \text{s.t. } \mathrm{V} \in \mathbb{R}^{\mathrm{n}\times\mathrm{r}}. \tag{8}$$

Whereas the rank function $\text{rank}(VV^\top)$ counts the number of nonvanishing eigenvalues of the similarity matrix $S = VV^\top$, the nuclear norm $\|VV^\top\|_*$ is the sum of the eigenvalues of the low rank and PSD similarity matrix $S = VV^\top$, which can be regarded as a proxy of the rank function.

However, calculating the gradient of the nuclear norm involves performing the SVD operation [57], [71] on the similarity matrix S, leading to a high computational cost when the data scale is large. To avoid this, Lemma 3.5 provides the tight upper bound of the nuclear norm to enable an efficient and comparable result [22]:

Lemma 3.5: $\|VV^\top\|_* \leq \|V\|_F^2 = \sum_i \|V_i\|^2$, where V_i is the i-th row-vector of V.

Proof: Following the Cauchy-Schawartz inequality [72], [73], for any real square matrices C and D, we have:

$$\begin{aligned}\text{tr}(CD) &= \sum_{ij} C_{ij} D_{ij} \leq \left(\sum_{ij} C_{ij}^2\right)^{\frac{1}{2}} \left(\sum_{ij} D_{ij}^2\right)^{\frac{1}{2}} \\ &= \left(\text{tr}(C^\top C)\right)^{\frac{1}{2}} \left(\text{tr}(D^\top D)\right)^{\frac{1}{2}} = \|C\|_F \|D\|_F.\end{aligned}$$

Recall that the nuclear norm $\|S\|_*$ is the sum of the eigenvalue of S [22]. We perform the SVD operation on the real symmetric and PSD similarity matrix S as $S = U\Sigma U^\top$, where Σ is the diagonal matrix of the singular values of S, and U is the associated orthogonal matrix [19] ($UU^\top = I$). Then

$$\begin{aligned}\|S\|_* &= \text{tr}(\Sigma) = \text{tr}(U^\top VV^\top U) = \text{tr}(UU^\top VV^\top) \\ &\leq \|V\|_F \|V\|_F \leq \frac{1}{2}(\|V\|_F^2 + \|V\|_F^2) = \|V\|_F^2\end{aligned}$$

The first inequality $\|VV^\top\|_* \leq \|V\|_F^2$ is based on the Cauchy-Schwarz inequality and the fact that U is an orthogonal matrix. The equality holds when $V = U\sqrt{\Sigma}$. Then, the equality $\|V\|_F^2 = \sum_i \|V_i\|^2$ holds, which follows directly from the definition of the Frobenius norm. □

C. Efficient Low Rank SMC Algorithm

1) SMCFN: SMC With Frobenius Norm as Regularizer: As Lemma 3.5 states that the nuclear norm $\|VV^\top\|_*$ is upper bounded by $\|V\|_F^2$, which allows the use of $\|V\|_F^2$ as a surrogate regularizer, thereby avoiding the high computation of the nuclear norm.

Algorithm 1: SMCFN: SMC with Frobenius Norm as Regularizer.

Input:
- $V^0 \in \mathbb{R}^{n\times r}$: Randomly initialized factorized matrix V^0 with rank r;
- γ: Stepsize;
- λ: Weighted parameter;
- T: Total iterations.

Output:
- $\hat{V} \in \mathbb{R}^{n\times r}$: Estimated factorized matrix;
- $\hat{S} \in \mathbb{R}^{n\times n}$: Estimated similarity matrix $\hat{S} = \hat{V}\hat{V}^\top$.

1: Define $F_1(V)$ shown as (9);
2: **for** t = 1,..., T **do**
3: Calculate gradient $\nabla_V F_1(V) = 4(VV^\top - S^0)V + 2\lambda V$;
4: Update $V^t = V^{t-1} - \gamma \nabla_V F_1(V^{t-1})$;
5: **end for**
6: Calculate $\hat{S} = \hat{V}\hat{V}^\top$;
7: **return** $\hat{S}$.

First, we leverage the Lemma 3.5 to replace the nuclear norm $\|VV^\top\|_*$ by the Frobenius norm $\|V\|_F^2$. Thus, problem (7) can be reformulated as:

$$\min_V F_1(V) := \|VV^\top - S^0\|_F^2 + \lambda\|V\|_F^2 \quad \text{s.t. } \mathrm{V} \in \mathbb{R}^{\mathrm{n}\times\mathrm{r}}, \tag{9}$$

where $S = VV^\top$, $S_{ij} = \{VV^\top\}_{ij}$, regularizer $R(V) = \|V\|_F^2$, $\lambda > 0$ is the weighted parameter.

Note that, though the MF technique [74], [75], [76], [77] suggests that the weighting parameter λ of the regularizer can be set to 0, which leads to a smooth function with simpler gradients. However, it is not suitable for the SMC problem setting, which aims to strictly ensure a lower-rank solution. This is the clear difference between the conventional MF technique for the MC problem and the low-rank SMC framework.

We further provide a general and efficient algorithm to solve problem (9) and summarize the entire process in Algorithm 1.

Line 1 shows the definition of the objective function $F_1(V)$. Lines 2-5 show the learning process for estimating the factorized matrix $\hat{V}$. Lines 6-7 show the calculation and return the similarity matrix $\hat{S}$.

Convergence Guarantee for Algorithm 1: We prove that Algorithm 1 can efficiently converge to a desirable solution and effectively minimize the objective function $F_1(V)$.

Theorem 3.6 (Convergence Guarantee): Taking $\gamma = \frac{1}{6G_1^2+\lambda}$, Algorithm 1 has:

$$\min_t ||\nabla_{V^t} F_1(V^t)||^2 \leq O\left(\frac{1}{T}\right)$$

Theorem 3.6 provides theoretical evidence that Algorithm 1 can effectively minimize the objective function $F_1(V)$ and converge to a desirable solution efficiently. It states that the convergence rate achieved $O(\frac{1}{T})$ with a fixed stepsize $\gamma = \frac{1}{6G_1^2+\lambda}$. As the number of iterations T increases, the objective $F_1(V)$ approaches the minimal value. Meanwhile, the convergence rate $O(\frac{1}{T})$ indicates that the objective square loss value decreases at a rate inversely proportional to the number of iterations T. (Detailed proof shown in Supplemental material B.)

2) SMCRN: SMC With Row-Wise Frobenius Norm as Regularizer: In SMCFN, while computational efficiency is ensured, the nuclear norm is relaxed from a hard constraint to a soft constraint. This relaxation potentially introduces a bias in estimation accuracy. Inspired by [37], [74], we extend the global optimality guarantee in the general MC framework to our SMC framework. Specifically, we delve into analyzing the bounds of the nuclear norm and provide an estimation guarantee for the novel low-rank SMC framework.

Recalling the problem (8), we adopt the nuclear norm to replace the rank function in problem (4) [27], [41], [70]. To enforce the optimality, we impose a stricter constraint by assuming that the nuclear norm $\|VV^\top\|_*$ is bounded by a constant c as a hard constraint. Then, the problem (4) can be reformulated as:

$$\|VV^\top - S^0\|_F^2 \quad \text{s.t. } \|\mathrm{VV}^\top\|_* \leq \mathrm{c},\ \mathrm{V} \in \mathbb{R}^{\mathrm{n}\times\mathrm{r}} \tag{10}$$

where the constant $c > 0$. Following Lemma 3.5, minimizing the nuclear norm can be transferred to minimizing the Frobenius norm of each row vector V_i sequentially. By using this formulation, we can extend the usage of regularizer from the general MC framework [37] into our SMC framework, which ensures the optimality guarantee, i.e., claims all the local optima are global optima. As a result, the low-rank SMC framework with optimal guarantee is converted to:

$$\min_V F_2(V) := \|VV^\top - S^0\|_F^2 \quad \text{s.t. } \|\mathrm{V_i}\| \leq \frac{\mathrm{c}}{\mathrm{n}}, \forall \mathrm{i} \tag{11}$$

To solve problem (11) smoothly [57], [66], [67], [68], [69], [70], we change a hard constraint to a soft constraint via the regularizer:

$$\min_V F_2(V) := \|VV^\top - S^0\|_F^2 + \lambda \sum_i^n (\|V_i\| - \alpha)_+^4 \tag{12}$$

where $\alpha := \frac{c}{n}$. Consistent with [37], we adopt a 4th power regularizer instead of a 2nd power regularizer primarily to ensure the Lipschitz 2nd order derivative.

Similarly, we provide a general and efficient gradient-based algorithm to solve problem (12), where the gradient is:

$$\nabla_V F_2(V) = 2(VV^\top - S^0)V + 4\lambda \sum_i^n (\|V_i\| - \alpha)_+^3 \frac{V_i}{\|V_i\|} \tag{13}$$

We summarize the entire process in Algorithm 2. Line 1 shows the objective function $F_2(V)$. Lines 2-5 show the learning process of estimating the factorized matrix $\hat{V}$. Lines 6-7 show the calculation and return the estimated similarity matrix $\hat{S}$.

In summary, our proposed SMCFN/SMCRN only requires to estimate the factorized matrix $\hat{V}$ instead of the entire matrix $S \in \mathbb{R}^{n\times n}$ by a general solver, where the computation cost is $O(n^2 r)$ instead of $O(n^3)$ [19], as well as the storage cost is $O(nr)$ instead of $O(n^2)$. As a result, both the computation cost and storage cost are significantly reduced since $r < n$.

Optimality Guarantee for Algorithm 2: Theorem 3.7 shows the SMCFN/SMCRN enjoys an optimality guarantee, which means all the local optima are global optima.

Algorithm 2: SMCRN: **SMC** with **R**ow-Wise Frobenius **N**orm as Regularizer.

Input:
 $V^0 \in \mathbb{R}^{n\times r}$: Randomly initialized factorized matrix V^0 with rank r;
 γ: Stepsize;
 λ: Weighted parameter;
 α: Threshold parameter;
 T: Total iterations.

1: Define $F_2(V)$ shown as (12);
2: **for** t = 1,..,T **do**
3: Calculate gradient $\nabla_V F_2(V) = 2(VV^\top - S^0)V + 4\lambda \sum_i^n (\|V_i\| - \alpha)_+^3 \frac{V_i}{\|V_i\|}$;
4: Update $V^t = V^{t-1} - \gamma \nabla_V F_2(V^{t-1})$;
5: **end for**
6: Calculate $\hat{S} = \hat{V}\hat{V}^\top$;
7: **return** $\hat{S}$.

Theorem 3.7 (Optimality Guarantee): Taking $\alpha^2 = O(\frac{1}{n})$, $\lambda = O(n)$, $\hat{S}$ solved by Algorithm 2 has optimal solution for (4) with probability $\Omega(\frac{\log n^2}{n^2})$ and

$$\|\hat{S} - S^*\| \leq \sigma_1^2 r \left\| O\left(G_2^2 nr \log n + \sqrt{pNrG_2^2 G_3^2 \log n}\right)\right\|^2 + pn^2.$$

Proof: According to Theorem 12 in [78], with $p \geq \Omega(\frac{\log v}{v})$ $\alpha^2 = O(\frac{1}{n})$ and $\lambda = O(n)$, we have V^∞ has the optimal solution for (4), where d denotes the dimension, n denotes total number of data samples, ρ denotes the missing ratio.

Since in our setting, $\rho = 1$, $v = n^2$, the desired result that Algorithm 2 has optimal solution for (4) with probability is obtained.

Since $\hat{S} := (V^\infty)(V^\infty)^\top$ is the optimal solution of (4), where V^∞ is the return of Algorithm 2. Then, by Theorem 3.4, we have the desired result. □

Theorem 3.8 (Convergence Guarantee for SMCRN): Taking $\gamma = \frac{1}{6G_1^2 + 2\lambda G_1^2}$, Algorithm 2 has:

$$\min_t \|\nabla_{V^t} F_2(V^t)\|^2 \leq O\left(\frac{1}{T}\right).$$

Theorem 3.8 provides theoretical evidence that Algorithm 2 can effectively minimize the objective function $F_2(V)$ and converge to a desirable solution efficiently. It states that the convergence rate achieved $O(\frac{1}{T})$ with a fixed stepsize $\gamma = \frac{1}{6G_1^2 + 2\lambda G_1^2}$. As T increases, the objective $F_2(V)$ approaches the minimal value. The key observation is that the 4th power regularizer $(\|V_i\| - \alpha)_+^4$ ensures Lipschitz continuous gradient, which is essential for the convergence analysis. (Detailed proof shown in Appendix C (available online).)

IV. Experiment

We perform a series of experiments on various real-world datasets, using a PC with an Intel i9-11900 2.5 GHz CPU and 32 GB of memory. We report the performance by repeating the comparison methods for 5 runs. Our evaluation aims to address the following research questions (RQ)[2]**RQ1**: How does our proposed SMCFN/SMCRN perform on SMC tasks and similarity search tasks? (IV-B)**RQ2**: How efficient is our proposed SMCFN/SMCRN? (IV-C)**RQ3**: How well are the low-rank matrix properties preserved in the proposed SMCFN/SMCRN? (IV-D)**RQ4**: How is the stability of our SMCFN/SMCRN? (IV-E)**RQ5**: How do hyperparameters affect our SMCFN/SMCRN? (IV-F)

A. Experimental Settings

• *Datasets 1) ImageNet* [79]: an image dataset contains 12M samples with 1,000 dimensions. *2) MNIST* [80]: an image dataset of handwritten digits (e.g., 0-9) contains 60 K samples with 784 dimensions. *3) CIFAR-10* [81]: an image dataset of 60K samples with 1,024 dimensions. *4) PROTEIN* [82]: a standard structural bio-informatics dataset contains 23K samples with 357 dimensions. *5) GoogleNews* [83]: a textual dataset of 3 M words in 300 dimensions.

• *Baseline Methods:* For all comparison methods in the experiments, we use public codes unless they are not available.

i) MDI methods. 1) Mean [48]: The missing values are imputed by the mean value in search data samples. *2) LR* [51]: The missing values are imputed by the multivariate linear regression between observed features and missing features. *3) GR* [84]: The missing values are imputed based on the low-rank assumption on the data matrix. *4) KFMC* [55]: The missing values are imputed and optimized in online data based on the high-rank assumption on the data matrix.

*ii) MC methods***.** *5) BPMF* [29]: The incomplete matrix is recovered by investigating the hidden patterns in the data using Bayesian inference. *6) FGSR* [30]: The incomplete matrix is recovered using a factored group-sparse regularizer without SVD operation. *7) NNFN* [40]: The incomplete matrix is recovered by a proximal algorithm based on non-negative SVD operation. *8) facNNFN* [40]: The scalable variant facNNFN recovered the incomplete matrix with a gradient descent algorithm. *9) PRGD* [42]: The similarity matrix is estimated by preconditioned Riemannian gradient descent. *10) NNSR* [85]: NNSR performs matrix completion using a nonconvex and nonsmooth sparse regularizer to handle noisy or corrupted observations.

*iii) SMC methods***.** *11) DMC* [15]: The similarity matrix is estimated by searching the approximated matrix with PSD based on Dykstra's alternating projection method. *12) CMC* [14]: The estimated similarity matrix is approximated by the cyclic projection with PSD constraint. *13) FMC* [39]: The estimated similarity matrix is approximated by estimating the similarity vector sequentially with PSD constraint.

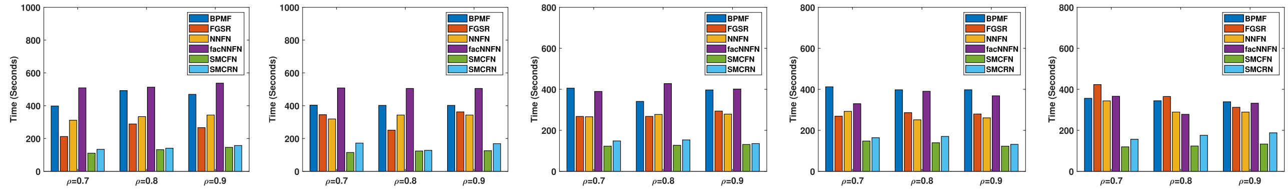
iv) Similarity Search Metric. nDCG: To evaluate the ranking quality by comparing the positions of relevant items in its output to a ground-truth ordering, we calculate the Normalized Discounted Cumulative Gain (*nDCG*) [87]: $nDCG_k = \frac{\mathrm{DCG}_k(\hat{S})}{\mathrm{IDCG}_k(S^*)}$, where $\mathrm{DCG}_k = \sum_{i=1}^{k} \frac{2^{\mathrm{rel}(\hat{S}_i)}-1}{\log_2(i+1)}$ with $\mathrm{rel}(\hat{S}_i)$ is the relevance score of the i-th item in $\hat{S}$ based on the positions in S^*, and $\mathrm{IDCG}_k = \sum_{i=1}^{k} \frac{2^{\mathrm{rel}(S^*)}-1}{\log_2(i+1)}$ with $\mathrm{rel}(S^*_i)$ is the relevance score of the i-th item in S^*. Note, when $\mathrm{IDCG}_k(S^*)$ is 0, then $\mathrm{DCG}_k(\hat{S})$ is 0, indicating non relevant items in top$k$ positions. nDCG varies in [0,1], comparing the ranking quality of the estimated $\hat{S}$ against

TABLE II

COMPARISONS OF RMSE, RECALL@TOP20%, NDCG@TOP20% WITH $n_q = 1,000$ SEARCH CANDIDATES AND $n_p = 200$ QUERY ITEMS, WHERE MISSING RATIO $\rho = \{0.7, 0.8, 0.9\}$, RANK $r = 100$, $\lambda = 0.001$, $\gamma = 0.001$, AND $T = 1,000$. $*$ THE BEST PERFORMANCE IS HIGHLIGHTED IN BOLD WHILE THE SECOND-BEST PERFORMANCE IS UNDERLINED. ↑ (RESP. ↓) INDICATES HIGHER (RESP. LOWER) IS BETTER.

Dataset/Method			0.7			0.8			0.9		
			RMSE↓	Recall↑	nDCG↑	RMSE↓	Recall↑	nDCG↑	RMSE↓	Recall↑	nDCG↑
ImageNet	**MDI**	**MEAN** [86]	2.6320	0.5878	0.7877	2.5412	0.5788	0.6819	2.5320	0.6128	0.8764
		LR [51]	1.5433	0.5984	0.7894	1.8786	0.6199	0.6826	2.0123	0.6195	0.8785
		GR [84]	1.1954	0.5874	0.7816	1.6456	0.6254	0.6838	1.9581	0.6251	0.8985
		KFMC [55]	1.2431	0.6014	0.7875	1.4513	0.6854	0.6656	1.1260	0.6764	0.9012
	MF	**FGSR** [30]	1.0450	0.1460	0.7809	1.3353	0.6475	0.6807	1.2592	0.6725	0.9242
		BPMF [29]	1.0341	0.1990	0.7548	1.4133	0.6239	0.6548	1.5633	0.6675	0.9257
	PSD	**CMC** [15]	0.7731	0.5980	0.7970	0.5946	0.6515	0.6436	0.5241	0.7614	0.8954
		DMC [14]	0.8329	0.6029	0.7548	0.6585	0.6990	0.6548	0.5136	0.7455	0.8959
		FMC [39]	0.6313	0.6772	0.6995	0.5665	0.5618	0.6995	0.5664	0.6715	0.8949
	LoR	**NNFN** [40]	1.6631	0.5940	0.7942	1.0880	0.5035	0.6139	0.5204	0.7621	0.9069
		facNNFN [40]	8.6170	0.4115	0.7931	1.0670	0.5580	0.6923	6.6200	0.2180	0.9165
		PRGD [42]	1.2843	0.5124	0.8017	0.9891	0.5783	0.6452	0.6821	0.5108	0.8125
		NNSR [85]	1.9200	0.5200	0.7300	1.3500	0.4400	0.5400	0.7100	0.6800	0.8200
	Ours	**SMCFN**	0.5861	0.7435	0.7974	0.4885	0.7160	0.6955	0.5049	**0.7814**	**0.9255**
		SMCRN	**0.4179**	**0.7928**	**0.8724**	**0.3430**	**0.7261**	**0.7060**	**0.3868**	0.7417	0.8143
MNIST	**MDI**	**MEAN** [86]	2.5918	0.6724	0.8279	2.5189	0.6773	0.5892	3.5188	0.6890	0.7298
		LR [51]	1.8247	0.6768	0.8875	1.8246	0.7184	0.5821	1.1295	0.6722	0.7391
		GR [84]	1.4761	0.6799	0.8765	1.6732	0.7199	0.6123	1.5314	0.6982	0.7572
		KFMC [55]	1.2482	0.6420	0.8855	1.4758	0.7156	0.6129	1.2751	0.6578	0.7866
	MF	**FGSR** [30]	0.9182	0.6338	0.8379	0.7470	0.6219	0.6379	1.1827	0.6574	0.7681
		BPMF [29]	0.9430	0.6219	0.8379	0.7347	0.6239	0.6379	1.2876	0.6879	0.7989
	PSD	**CMC** [15]	0.9887	0.6284	0.8622	0.7288	0.6244	0.6123	0.5214	0.7314	0.8248
		DMC [14]	1.0430	0.5893	0.6379	0.5747	0.6309	0.6379	0.5145	0.7251	0.8483
		FMC [39]	0.7459	0.5775	0.6977	0.6509	0.5775	0.5995	0.5269	0.7274	0.7977
	LoR	**NNFN** [40]	1.6631	0.5940	0.7942	1.0880	0.5035	0.6139	0.5204	0.7621	0.9069
		facNNFN [40]	8.6170	0.4115	0.7931	1.0670	0.5580	0.6923	6.6200	0.2180	0.9165
		PRGD [42]	0.9392	0.4876	0.7724	0.9114	0.5529	0.6198	0.6763	0.4852	0.7841
		NNSR [85]	1.2568	0.4987	0.7863	0.9752	0.5641	0.6319	0.5894	0.4975	0.7982
	Ours	**SMCFN**	0.7044	0.6832	**0.8903**	0.5097	0.6822	0.6878	**0.4239**	0.7491	0.8658
		SMCRN	**0.4180**	**0.7611**	0.8128	**0.4602**	**0.7596**	**0.8178**	0.4317	**0.8165**	**0.8678**
PROTEIN	**MDI**	**MEAN** [86]	3.1196	0.6989	0.7813	2.7642	0.6876	0.6838	2.4871	0.6343	0.8275
		LR [51]	1.8728	0.6878	0.7813	1.4760	0.6856	0.6848	1.7364	0.6453	0.8173
		GR [84]	1.7676	0.7184	0.7839	1.2315	0.6896	0.6865	1.6563	0.6275	0.8757
		KFMC [55]	1.8735	0.6871	0.7990	1.1285	0.6013	0.6864	1.3458	0.7187	0.8793
	MF	**BPMF** [29]	1.2793	0.6165	0.7962	1.5498	0.6105	0.6962	1.0149	0.7265	0.8476
		FGSR [30]	1.3123	0.6340	0.7950	1.2130	0.6134	0.6950	1.5372	0.6983	0.7983
	PSD	**CMC** [15]	0.8085	0.6175	0.7989	0.5776	0.6138	0.6985	0.7141	0.7241	0.8848
		DMC [14]	0.9477	0.6340	0.7950	0.5231	0.6134	0.6950	0.5645	0.6600	0.7984
		FMC [39]	0.6109	0.7075	0.7998	0.5362	0.6743	0.6843	0.5322	0.6743	0.6949
	LoR	**NNFN** [40]	1.4627	0.6470	0.7954	2.0510	0.6144	0.6953	0.6322	0.7289	0.8731
		facNNFN [40]	1.4986	0.6540	0.7964	1.6579	0.6085	0.6961	3.2316	0.5988	0.7843
		PRGD [42]	0.9705	0.5053	0.7936	0.9627	0.5704	0.6387	0.6942	0.5031	0.8059
		NNSR [85]	1.6729	0.6038	0.7583	2.3567	0.5714	0.6589	0.7845	0.6721	0.8156
	Ours	**SMCFN**	0.5553	0.7366	0.8170	**0.4241**	0.7129	0.6980	0.5115	0.7685	**0.8922**
		SMCRN	**0.4227**	**0.7799**	**0.8653**	0.4864	**0.7296**	**0.7826**	**0.4185**	**0.8281**	0.8340
CIFAR10	**MDI**	**MEAN** [86]	6.5890	0.5511	0.6806	6.3863	0.4731	0.6783	9.6703	0.5938	0.7988
		LR [51]	1.9312	0.7459	0.7008	1.6657	0.4728	0.6727	3.0418	0.6036	0.7995
		GR [84]	2.7911	0.7314	0.7076	1.0575	0.4714	0.6736	2.0572	0.5125	0.7419
		KFMC [55]	1.5571	0.7417	0.7051	1.1822	0.4273	0.6781	3.0500	0.5042	0.7474
	MF	**FGSR** [30]	0.5980	0.7092	0.7020	0.5742	0.5150	0.6320	1.3159	0.5333	0.7982
		BPMF [29]	0.7432	0.7313	0.7020	0.5920	0.5340	0.6320	1.4170	0.5625	0.7992
	PSD	**CMC** [15]	0.6539	0.7276	0.6772	0.5018	0.5280	0.5739	0.5462	0.7004	0.7988
		DMC [14]	0.5980	0.6192	0.6320	0.4742	0.5915	0.6320	0.7714	0.6771	0.8290
		FMC [39]	0.6269	0.6644	0.6984	0.7347	0.4688	0.5984	0.7365	0.6613	0.8430
	LoR	**NNFN** [40]	0.7795	0.7290	0.6251	0.7190	0.5220	0.6868	0.8352	0.5688	0.8275
		facNNFN [40]	2.0373	0.6280	0.7079	1.7810	0.5213	0.6944	0.8305	0.7167	0.8288
		PRGD [42]	0.9481	0.4912	0.7798	0.9383	0.5592	0.6264	0.6837	0.4903	0.7912
		NNSR [85]	0.9376	0.6845	0.5829	0.8732	0.4761	0.6458	1.0053	0.5109	0.7732
	Ours	**SMCFN**	0.5689	**0.8223**	0.7866	**0.4443**	0.5951	0.6945	**0.4546**	0.7708	**0.8951**
		SMCRN	**0.5121**	0.7708	**0.8563**	0.5213	**0.7083**	**0.7533**	0.5209	**0.7827**	0.7901
GoogleNews	**MDI**	**MEAN** [86]	2.4314	0.6985	0.7821	2.1510	0.5420	0.7124	1.9898	0.5715	0.7851
		LR [51]	1.2130	0.6985	0.7941	1.2145	0.5983	0.7154	1.6743	0.5568	0.8012
		GR [84]	1.2140	0.6741	0.7214	1.7530	0.5012	0.7256	1.5438	0.5939	0.7987
		KFMC [55]	1.1065	0.6784	0.7531	1.5410	0.5214	0.7050	1.4785	0.6136	0.7989
	MF	**FGSR** [30]	1.0774	0.6103	0.4409	0.9451	0.5103	0.7409	1.7516	0.6215	0.8125
		BPMF [29]	2.2310	0.6025	0.2393	0.9432	0.5025	0.7409	1.4817	0.6255	0.8254
	PSD	**CMC** [15]	1.0000	0.6285	0.7095	0.9962	0.5525	0.7291	0.7788	0.6604	0.8314
		DMC [14]	1.0774	0.6025	0.7409	0.7513	0.5103	0.7409	0.5378	0.6617	0.8125
		FMC [39]	0.8491	0.6840	0.7998	0.6559	0.5645	0.7065	0.6573	0.6451	0.7977
	LoR	**NNFN** [40]	0.9055	0.7263	0.7904	0.7488	0.5247	0.7291	0.8058	0.6627	0.8542
		facNNFN [40]	2.2081	0.7227	0.7502	1.7425	0.5215	0.7540	3.1549	0.6613	0.8235
		PRGD [42]	0.9624	0.5089	0.7891	0.9595	0.5678	0.6351	0.6912	0.5067	0.8003
		NNSR [85]	1.0734	0.6721	0.7358	0.9157	0.4736	0.6729	0.9785	0.5943	0.7841
	Ours	**SMCFN**	0.8233	0.7762	0.8488	0.7080	**0.6432**	**0.7845**	0.4893	0.6961	0.8854
		SMCRN	**0.4108**	**0.8378**	**0.8539**	**0.4211**	0.6233	0.6498	**0.4227**	**0.8285**	**0.9217**

[2]Code available at: https://drive.google.com/drive/folders/1kXVdYJyKS0TJvZ-mqdVOexTziG6qupv_?usp=drive_link



(a) ImageNet, r=100 (b) MNIST, r=100 (c) PROTEIN, r=100 (d) CIFAR-10, r=100 (e) GoogleNews, r=100

Fig. 1. Running Time versus ρ on ImageNet dataset, with $n_p = 1,000$ search candidates and $n_q = 200$ query items, with rank $r = \{50, 100, 200\}$, $\lambda = 0.001$, $\gamma = 0.001$, $T = 1,000$. (Best viewed in $\times$ 2 sized color pdf file).

the ground truth S^* up to a certain rank position k. A larger nDCG denotes better performance.

• **Implementation details**, **parameters setting**, and **evaluation metric** are in Appendix E (available online).

B. Effectiveness Evaluation

Table II shows the results of RMSE, Recall@top 20%, and nDCG@top20% for all comparison methods. Considering SMC tasks, we find our SMCFN/SMCRN consistently yielded the best similarity matrix completion performance compared with other baseline methods in terms of RMSE, which is consistent with the reliable estimated guarantee shown in Theorem 3.1. In terms of similarity search tasks, SMCFN/SMCRN also yields the best Recall and nDCG compared to the comparison methods, revealing that the estimated similarity matrix can preserve the pairwise similarity well and benefit the downstream search tasks.

C. Efficiency Evaluation

Fig. 1 shows the running time varying with missing ratio ρ on various datasets of the comparison methods. Since our SMCFN/SMCRN adopts the efficient MF technique, we only report the running time with the comparison methods that involve MF techniques. Among all the comparison methods, our SMCFN/SMCRN achieves the highest efficiency, which is consistent with our motivation to design an efficient low-rank SMC framework. Combining the effectiveness results in Section IV-B, our SMCFN/SMCRN achieves the best SMC performance with the highest efficiency, showcasing the superiority of our Low-Rank SMC framework. (Complexity Analysis shown in Supplemental material A.)

D. Lower-Rank Matrix Property Verification

Table III shows the rank of the estimated matrix $\hat{S}$ of comparison methods.

Specifically, the rank is calculated by forcing all the eigenvalues larger than a threshold value 10^{-15}, which is the default setting in MATLAB. Our SMCFN/SMCRN ensures that the *Lower-Rank Matrix Property* always holds, which verifies the theoretical observations in Theorem 3.3. Combining the theoretical analysis and empirical results, we conclude that SMCFN/SMCRN achieves a satisfying estimation performance with high efficiency.

TABLE III
RANK($\hat{S}$), WITH $n_p = 1,000$ SEARCH CANDIDATES AND $n_q = 200$ QUERY ITEMS, WHERE $r = 100$ WITH VARIOUS MISSING RATIO ρ, FIXED $\lambda = 0.001$, $\gamma = 0.001$, AND $T = 1,000$

Datasets	ρ	S^*	CMC	DMC	FMC	FGSR	BPMF	NNFN	facNNFN	SMCFN	SMCRN
ImageNet	**0.7**	**978**	1200	1200	1064	683	1052	1200	1200	**81**	**80**
	0.8	**992**	1200	1200	1104	691	1052	1200	1200	**80**	**89**
	0.9	**1000**	1200	1200	1156	951	1060	1200	1200	**77**	**98**
MNIST	**0.7**	**484**	1200	1200	1172	684	1068	1200	1200	**75**	**87**
	0.8	**501**	1200	1200	1168	699	1060	1200	1200	**71**	**95**
	0.9	**515**	1200	1200	1180	901	1068	1200	1200	**70**	**97**
PROTEIN	**0.7**	**357**	1200	1200	1184	684	1044	1200	1200	**78**	**80**
	0.8	**357**	1200	1200	1196	692	1048	1200	1200	**77**	**92**
	0.9	**357**	1200	1200	1200	933	1044	1200	1200	**82**	**99**
CIFAR-10	**0.7**	**501**	1200	1200	1200	684	1044	1200	1200	**76**	**80**
	0.8	**512**	1200	1200	1200	692	1048	1200	1200	**84**	**92**
	0.9	**512**	1200	1200	1200	952	1068	1200	1200	**78**	**94**
GoogleNews	**0.7**	**300**	1200	1200	1144	864	1080	1200	1200	**85**	**91**
	0.8	**300**	1200	1200	1157	904	1076	1200	1200	**83**	**91**
	0.9	**300**	1200	1200	1156	1040	1068	1200	1200	**79**	**97**

TABLE IV
DETAILED OF VARIOUS COMPARATIVE ABLATION METHODS. $*$ F-NORM DENOTES FROBENIUS NORM, N-NORM DENOTES NUCLEAR NORM.

Methods		LoR	PSD	Regularizer	Solver
Ablation Study	**SMC_F**	×	✓	F-Norm	Gradient Descent
	SMC_NR	×	✓	No Regularizer	Gradient Descent
	SMC_GD	✓	✓	N-Norm	Gradient Descent
	SMC_ON	✓	×	N-Norm	SVD Operation
	SMC_fON	✓	×	N-Norm	Gradient Descent
SMC	**CMC**[15]	×	✓	No Regularizer	Cyclic Projection
	DMC[14]	×	✓	No Regularizer	Dykstra's Projection
	FMC[39]	×	✓	No Regularizer	Lagrangian Method
Ours	**SMCFN**	✓	✓	F-Norm	Gradient Descent
	SMCRN	✓	✓	Row-Wise F-norm	Gradient Descent

E. Ablation Study

We further apply a series of ablation studies to better understand the importance of the specified lower-rank matrix property. Table IV shows the detailed comparison of various methods.

To keep consistent with our methods, all the ablation methods are implemented by the general gradient descent method, where the weighted parameter $\lambda = 0.001$, stepsize $\gamma = 0.001$, rank $r = 100$, and the maximum number of iterations $T = 1,000$.

i) SMC_F: Directly use the Frobenius norm as the regularizer.

$$\min_V \|VV^\top - S^0\|_F^2 + \lambda\|VV^\top\|_F^2.$$

ii) SMC_NR: Directly ignore the regularizer term.

$$\min_V \|VV^\top - S^0\|_F^2.$$

TABLE V
COMPARISON OF ABLATION METHODS ON VARIOUS VARIATIONS OF RMSE, RECALL@TOP20%, NDCG@TOP20% WITH $n_q = 1,000$ SEARCH CANDIDATES AND $n_q = 200$ QUERY ITEMS, WHERE MISSING RATIO $\rho = 0.7, 0.8, 0.9$, RANK $r = 100$, $\lambda = 0.001$, $\gamma = 0.001$, AND $T = 1,000$

Datasets	Method	0.7			0.8			0.9		
		RMSE↓	Recall↑	nDCG ↑	RMSE↓	Recall↑	nDCG ↑	RMSE↓	Recall↑	nDCG ↑
ImageNet	SMC_F	0.6311	0.6476	0.7050	0.5570	0.6459	0.6402	0.5746	0.7441	0.8424
	SMC_NR	0.6370	0.6503	0.7202	0.5252	0.6277	0.6214	0.5916	0.6953	0.9132
	SMC_GD	**0.3922**	0.6614	0.6932	0.3465	0.4518	0.5063	**0.3260**	0.5967	0.8303
	MC_ON	0.8101	0.7014	0.8493	1.1426	0.6035	0.5808	0.3967	0.7300	0.7228
	MC_fON	0.8592	0.7030	0.7962	0.5564	0.6696	0.6769	0.5568	0.5879	0.7670
	CMC [15]	0.7731	0.5980	0.7970	0.5946	0.6515	0.6436	0.5241	0.7614	0.8954
	DMC [14]	0.8329	0.6029	0.7548	0.6585	0.6990	0.6548	0.5136	0.7455	0.8959
	FMC [39]	0.6313	0.6772	0.6995	0.5665	0.5618	0.6995	0.5664	0.6715	0.8949
	SMCFN(Ours)	0.5861	0.7435	0.7974	0.4885	0.7160	0.6955	0.5049	**0.7814**	0.9255
	SMCRN(Ours)	0.4179	**0.7928**	**0.8724**	**0.3430**	**0.7261**	**0.7060**	0.3868	0.7417	0.8143
MNIST	SMC_F	0.7962	0.6124	0.8042	0.5407	0.6190	0.6520	0.4796	0.6503	0.8462
	SMC_NR	0.7291	0.5884	0.8378	0.5653	0.6485	0.6381	0.4656	0.6494	0.8624
	SMC_GD	0.4194	0.6179	0.7140	**0.4548**	0.4676	0.6509	0.4263	0.6216	0.7216
	MC_ON	0.5282	0.6080	0.6232	1.1294	0.5934	0.7911	0.7128	0.6593	0.8202
	MC_fON	0.6610	0.6047	0.6983	0.5258	0.6248	0.6290	0.6367	0.8033	0.8605
	CMC [15]	0.9887	0.6284	0.8622	0.7288	0.6244	0.6123	0.5214	0.7314	0.8248
	DMC [14]	1.0430	0.5893	0.6379	0.5747	0.6309	0.6379	0.5145	0.7251	0.8483
	FMC [39]	0.7459	0.5775	0.6977	0.6509	0.5775	0.5995	0.5269	0.7274	0.7977
	SMCFN(Ours)	0.7044	0.6832	**0.8903**	0.5097	0.6822	0.6878	**0.4239**	0.7491	0.8658
	SMCRN(Ours)	**0.4180**	**0.7611**	0.8128	0.4602	**0.7596**	**0.8178**	0.4317	**0.8165**	**0.8678**
PROTEIN	SMC_F	0.5976	0.6772	0.7746	0.4513	0.6555	0.6227	0.5542	0.7677	0.8592
	SMC_NR	0.5946	0.6766	0.7990	0.4974	0.6825	0.6556	0.5875	0.7112	0.8913
	SMC_GD	0.4871	0.6516	0.6150	0.5016	0.5077	0.6952	0.5040	0.5984	0.8404
	MC_ON	1.1457	0.5921	0.7192	0.9351	0.5409	0.7031	1.1322	0.6734	0.7769
	MC_fON	0.6834	0.6621	0.6716	0.6007	0.6191	0.7431	0.5345	0.7272	0.7769
	CMC [15]	0.8085	0.6175	0.7989	0.5776	0.6138	0.6985	0.7141	0.7241	0.8848
	DMC [14]	0.9477	0.6340	0.7950	0.5231	0.6134	0.6950	0.5645	0.6600	0.7984
	FMC [39]	0.6109	0.7075	0.7998	0.5362	0.6743	0.6843	0.5322	0.6743	0.6949
	SMCFN(Ours)	0.5553	0.7366	0.8170	0.4241	0.7129	0.6980	0.5115	0.7685	**0.8922**
	SMCRN(Ours)	**0.4227**	**0.7799**	**0.8653**	0.4864	0.7296	0.7826	0.4185	0.8281	0.8340
CIFAR-10	SMC_F	0.6087	0.7860	0.7368	0.5356	0.5055	0.5977	0.5306	0.6741	0.8695
	SMC_NR	0.6632	0.7594	0.7650	0.4907	0.5917	0.6632	0.5445	0.6915	0.8927
	SMC_GD	**0.4144**	0.5154	0.6180	0.4509	0.5773	0.5210	**0.4172**	0.6456	0.7855
	MC_ON	0.9238	0.6487	0.7479	0.6308	0.5833	0.6218	0.5947	0.6438	0.6196
	MC_fON	0.9423	0.5986	0.7529	0.8588	0.5161	0.5617	0.6535	0.7079	0.7785
	CMC [15]	0.6539	0.7276	0.6772	0.5018	0.5280	0.5739	0.5462	0.7004	0.7988
	DMC [14]	0.5980	0.6192	0.6320	0.4742	0.5915	0.6320	0.7714	0.6771	0.8290
	FMC [39]	0.6269	0.6644	0.6984	0.7347	0.4688	0.5984	0.7365	0.6613	0.8430
	SMCFN(Ours)	0.5689	**0.8223**	0.7866	**0.4443**	0.5951	0.6945	0.4546	0.7708	**0.8951**
	SMCRN(Ours)	0.5121	0.7708	**0.8563**	0.5213	**0.7083**	**0.7533**	0.5209	**0.7827**	0.7901
GoogleNews	SMC_F	0.8612	0.6553	0.7930	0.7754	**0.6598**	0.6790	0.6395	0.6771	0.7736
	SMC_NR	0.8603	0.6554	0.7930	0.7754	0.6370	0.6899	0.6394	0.6771	0.7736
	SMC_GD	0.4667	0.6678	0.7131	**0.4045**	0.5992	0.6619	**0.3913**	0.6361	0.6731
	MC_ON	0.6689	0.8376	0.8334	0.5129	0.5077	0.6338	0.7997	0.8000	0.8424
	MC_fON	0.6196	0.6551	0.7299	0.5320	0.5436	0.5894	0.4952	0.7244	0.8206
	CMC [15]	1.0000	0.6285	0.7095	0.9962	0.5525	0.7291	0.7788	0.6604	0.8314
	DMC [14]	1.0774	0.6025	0.7409	0.7513	0.5103	0.7409	0.5378	0.6617	0.8125
	FMC [39]	0.6109	0.7075	0.7998	0.6559	0.5645	0.7065	0.6573	0.6451	0.7977
	SMCFN(Ours)	0.8233	0.7762	0.8488	0.7080	0.6432	**0.7845**	0.4893	0.6961	0.8854
	SMCRN(Ours)	**0.4108**	**0.8378**	**0.8539**	0.4211	0.6233	0.6498	0.4227	**0.8285**	**0.9217**

iii) SMC_GD: The nuclear norm is applied to the estimated matrix generated from the factorized matrix.

$$\min_{V} \|VV^\top - S^0\|_F^2 + \lambda \|VV^\top\|_*.$$

i) SMC_ON: The original nuclear norm is directly used as the regularizer.

$$\min_{S} \|S - S^0\|_F^2 + \lambda \|S\|_* \quad \text{s.t.} \quad S \succeq 0.$$

i) SMC_fON: Matrix factorization is adopted instead of Cholesky factorization.

$$\min_{W,H} \|WH^\top - S^0\|_F^2 \quad \text{s.t.} \quad WH^\top \succeq 0.$$

Table V shows the performance of RMSE, Recall@top 20%, and nDCG@top20% of variations on various datasets. Among these different settings, our SMCFN/SMCRN outperforms the other variations on various datasets, which further reveals the necessity of forcing the similarity matrix to be low-rank by nuclear norm and finding the optimal results via well-defined, efficient, and general solvers. Combining this result with the performances of CMC, DMC, and FMC, we summarize that using either PSD or low-rank properties on SMC problems cannot yield the best performance.

F. Hyperparameter Sensitivity

We conduct a series of experiments to further analyze the sensitivity of SMCFN/SMCRN with learning-based methods. The parameters examined included the rank r, weighted parameter λ, and stepsize γ.

Effect of r: Fig. 2 shows the RMSE/Recall varying with various input rank $r = \{50, 100, 200\}$ on various datasets. It is obvious that the RMSE of SMCFN/SMCRN decreases with the increasing number of rank r from 50 to 100, and the Recall of SMCFN/SMCRN shows the opposite tendencies. Meanwhile, the performance slightly changes when the rank r increases from 100 to 200.

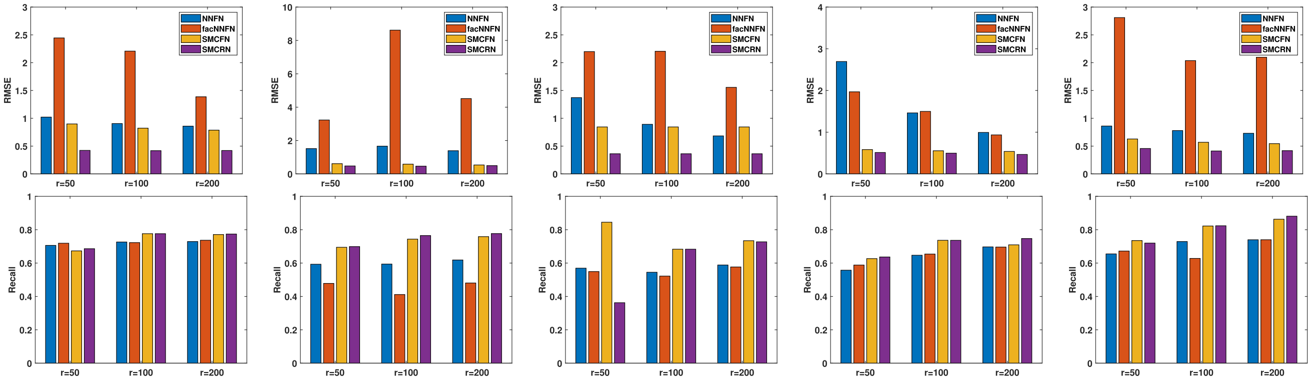
Effect of γ: Fig. 3 shows RMSE/Recall varying with weighted parameter λ on various datasets. Clearly, the RMSE/Recall of our SMCFN/SMCRN is always superior to the baseline methods. The performance slightly changes on MNIST and CIFAR datasets, but fluctuates on ImageNet, PROTEIN, and GoogleNews datasets, i.e., first decreases from $\gamma = 0.0001$ to $\gamma = 0.01$, and increases from $\gamma = 0.1$ to $\gamma = 10$. It is probably caused by the different data distribution across different datasets.

Effect of λ: Fig. 4 shows the RMSE/Recall varying with stepsize γ on various datasets. Similar to the analysis about γ, the RMSE of SMCFN/SMCRN also shows different performances on different datasets. Specifically, we find that both SMCFN/SMCRN perform stably on MNIST, PROTEIN, CIFAR, and GoogleNews datasets with a fixed λ and various γ. However, the performance is not so stable on the ImageNet datasets. It is probably caused by the large data size of ImageNet, where the underlying information is somehow difficult to capture. This tendency could be attributed to the substantial size of the ImageNet dataset, where capturing the underlying information should be done more carefully.

V. CONCLUSION AND FUTURE WORK

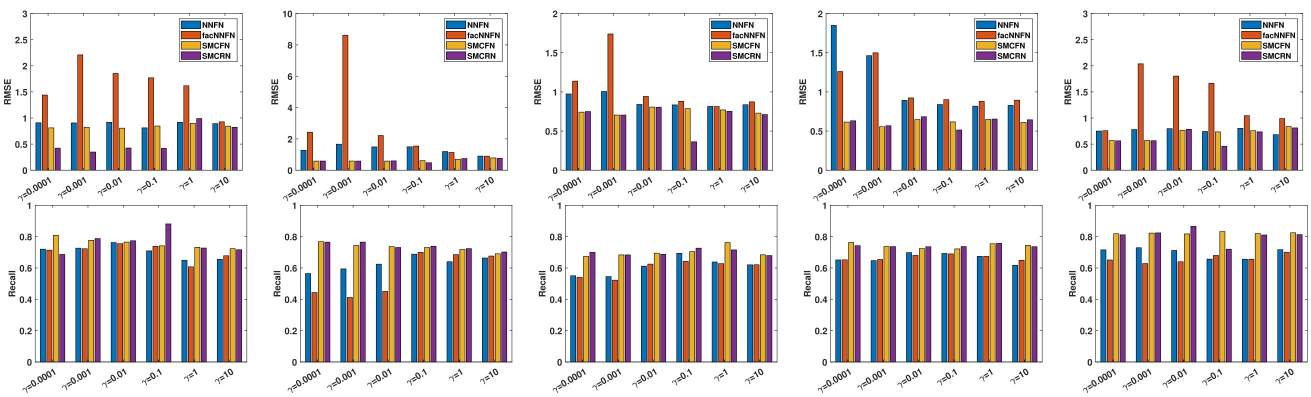
In this paper, we propose a novel SMC framework that integrates the symmetric, PSD, and low-rank properties, providing an accurate and efficient solution for similarity search tasks with data incompleteness. Furthermore, we introduce a lower-rank matrix property to enhance the optimality guarantees and accordingly develop two regularizers that seamlessly integrate into the SMC framework. Then, we propose two general and efficient SMCFN/SMCRN algorithms to implement this framework. Theoretical analysis and empirical evaluations support the effectiveness and efficiency of SMCFN/SMCRN, shedding new light on similarity search problems with data incompleteness in practical applications.

In the future, more efforts will be concentrated on developing more methods to approximate the SVD operation. Whereas the low-rank property is crucial for SMC problems, existing methods such as truncated SVD and nuclear norm minimization can still be computationally intensive, particularly when



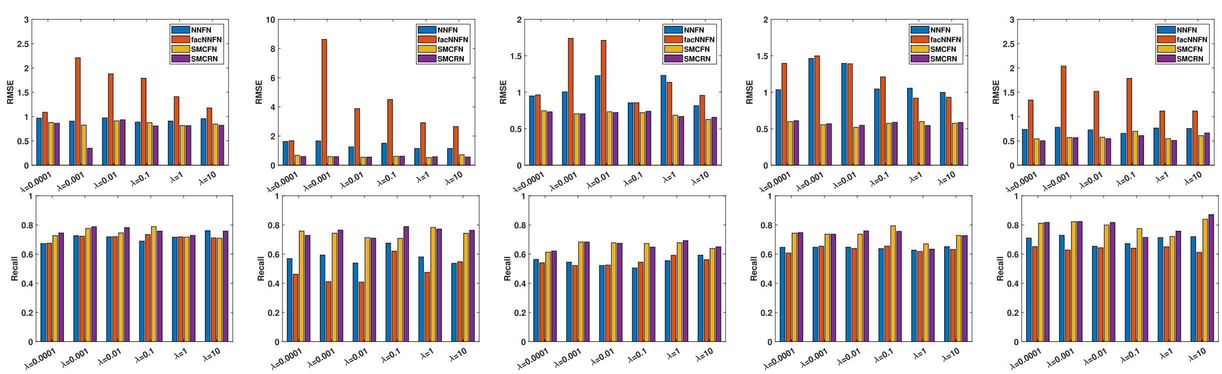
(a) ImageNet (b) MNIST (c) PROTEIN (d) CIFAR-10 (e) GoogleNews

Fig. 2. RMSE/Recall@20% versus rank r on various dataset, with $n_p = 1,000$ search candidates and $n_q = 200$ query items, where missing ratio $\rho = 0.7$, rank $r = 50, 100, 200$, $\lambda = 0.001$, $\gamma = 0.001$, iterations $T = 1,000$.



(a) ImageNet (b) MNIST (c) PROTEIN (d) CIFAR-10 (e) GoogleNews

Fig. 3. RMSE/Recall@top20% versus stepsize γ on various dataset, with $n_p = 1,000$ search candidates and $n_q = 200$ query items, where missing ratio $\rho = 0.7$, rank $r = 100$, $\lambda = 0.001$, iterations $T = 1,000$.



(a) ImageNet (b) MNIST (c) PROTEIN (d) CIFAR-10 (e) GoogleNews

Fig. 4. RMSE/Recall@top20% versus weighted parameter λ on various dataset, with $n_p = 1,000$ search candidates and $n_q = 200$ query items, where missing ratio $\rho = 0.7$, rank $r = 100$, $\gamma = 0.001$, iterations $T = 1,000$.

large-scale datasets are involved. To enhance the efficiency of SMC while maintaining its accuracy, future studies may consider investigating simpler and more efficient approximations of SVD.

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